Arimoclomol for BUB1B-MVA

Conditional preclinical repurposing proposal

Can this candidate restore functional BubR1?

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No drug rescue experiment has been performed.

Variant abundance and function

Phase is inferred. N1002K function remains unmeasured.

Protein amount

Expression, turnover and localisation

Protein function

Full-length WT and variant compared at matched abundance

Possible outcomes: abundance-limited, function-limited, mixed or indeterminate

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Four experimental gates

A **Establish the variant's effect**
Full-length protein at matched abundance

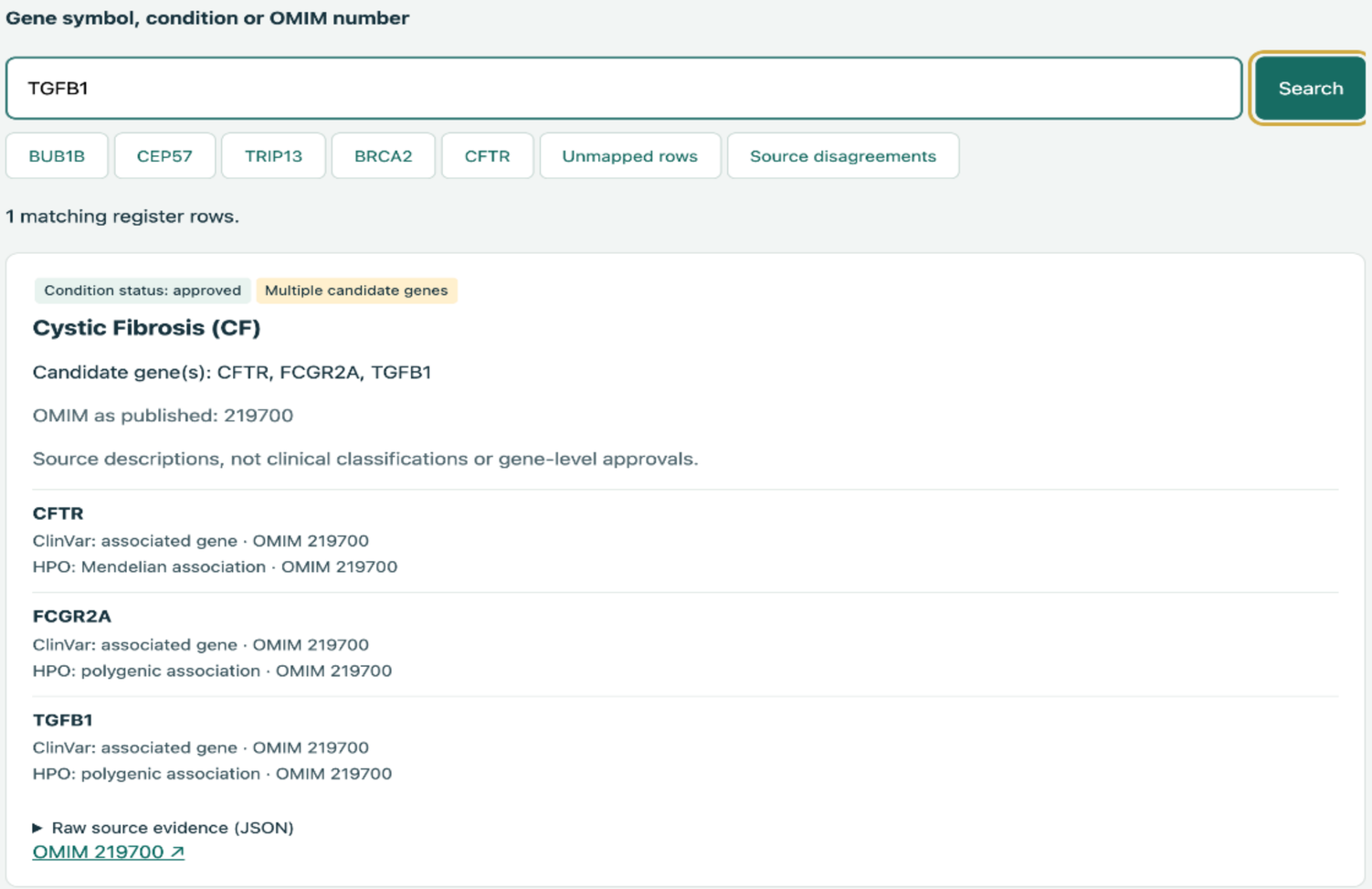
B **Validate measurement and response**
Qualified controls and measured exposure

C **Measure segregation and cell fate**
Separate division, death, arrest and selection

D **Apply the prespecified decision rules**
Confidence bounds separate advance, reject and inconclusive

Failed controls: indeterminate. Threshold-touching intervals: inconclusive.

Gene-to-condition navigator



28 queries

47 returned gene-row associations reviewed

Source roles visible

Shared-source audit
Clinical validity untested

Rules that reject misleading hits

Implemented rescue-decision prototype

40

frozen synthetic cases
matched expected decisions

Also: 6 real SpliceAI controls passed
9 injected checker faults rejected

Software verification. Biological rescue remains unmeasured.

The next experiment

Does N1002K retain function that arimoclomol can rescue?

Delivered now

Code, source trail and
a proposed laboratory protocol

Evidence still needed

Independent functional rescue
at relevant exposure